Aerial Data Processing in Agisoft Metashape

without Ground Control Points (GCPs)

1.0 Introduction to the GCP-less Workflow

A standard operating procedure for generating a Digital Elevation Model (DEM) and an orthomosaic from aerial imagery using Agisoft Metashape Professional. Specifically, this guide details the workflow to be used when Ground Control Points (GCPs) are not available. This methodology is strategically important for projects requiring rapid assessments, preliminary surveys, or for scenarios where ground access is restricted or impractical. The procedure detailed below follows a clear sequence of processing stages, each building upon the last to produce the final data products.

2.0 Workflow Overview

The end-to-end process for generating a DEM and orthomosaic without GCPs consists of four primary stages. These stages are executed sequentially within the Agisoft Metashape software environment.

- **Photo Alignment:** Calculates the position and orientation of each photo and generates an initial sparse point cloud.
- **Dense Point Cloud Generation:** Creates a detailed, high-density 3D point cloud of the surveyed area.
- **Digital Elevation Model (DEM) Generation:** Converts the unstructured 3D point cloud into a 2.5D raster surface representing terrain elevation.
- **Orthomosaic Generation:** Performs orthorectification on the source images, stitching them into a single, geometrically corrected, and uniformly scaled map.

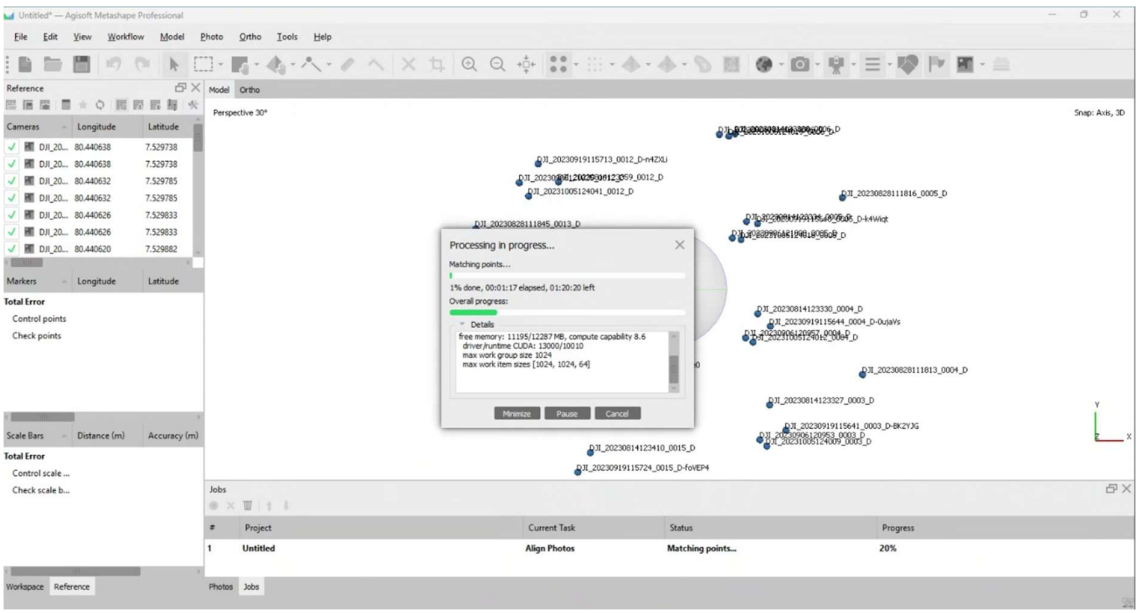
The following sections provide a detailed, step-by-step breakdown of each of these core processing stages.

3.0 Detailed Procedural Steps

3.1 Step 1: Photo Alignment

The Photo Alignment stage is the foundational step in the photogrammetric process. Its purpose is to analyze the entire set of aerial photos to identify common features, which are then used to calculate the precise position and orientation—its exterior orientation (X,Y,Z, ω, φ, κ) and interior camera parameters—for each image. This calculation results in the generation of an initial sparse point cloud that serves as the geometric framework for all subsequent steps.

The process is initiated by running the "Align Photos" task. As seen in the software interface below, the first phase of this task involves "Matching points" between overlapping images to find common tie points.



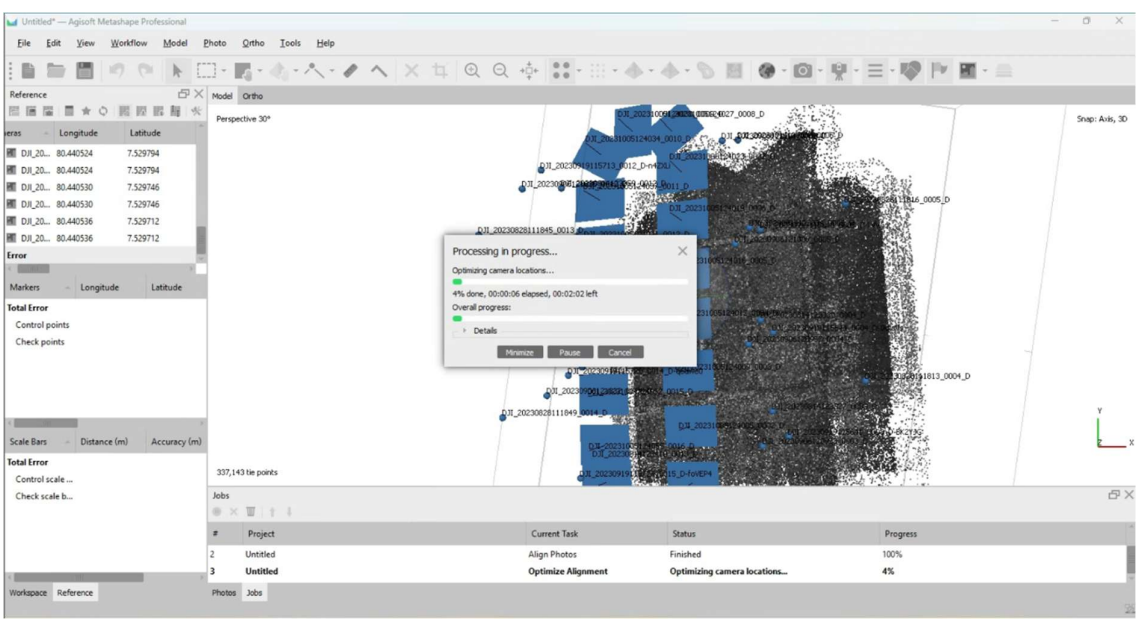


Figure 1: The "Align Photos" process is initiated, with the software matching common points across the image set. The background shows the initial estimated camera positions.

As shown in Figure 1, the **Jobs** panel displays "Align Photos" as the current task, while the progress dialog confirms the "Matching points..." status. This is the most computationally intensive initial step, as it involves feature detection across thousands of image pairs and a complex bundle adjustment calculation to solve for all camera parameters simultaneously. Upon completion, a sparse point cloud and the calculated camera positions form the basis for densifying the 3D data.

3.2 Step 2: Dense Point Cloud Generation

Following a successful photo alignment, the next strategic step is to generate a dense point cloud. This process uses the camera positions and the **337,143 tie points** from the sparse alignment as a geometric framework to generate millions of additional points, creating a comprehensive 3D model. This densification is preceded by an intermediate "Optimize Alignment" step, which refines the camera positions and lens distortion parameters to improve the overall geometric accuracy of the project.

The densification task is executed via the "Build Point Cloud" command. The "Generating depth maps..." status indicates the core of the densification algorithm, where the software computes depth from multiple camera perspectives for each pixel to build the 3D model.

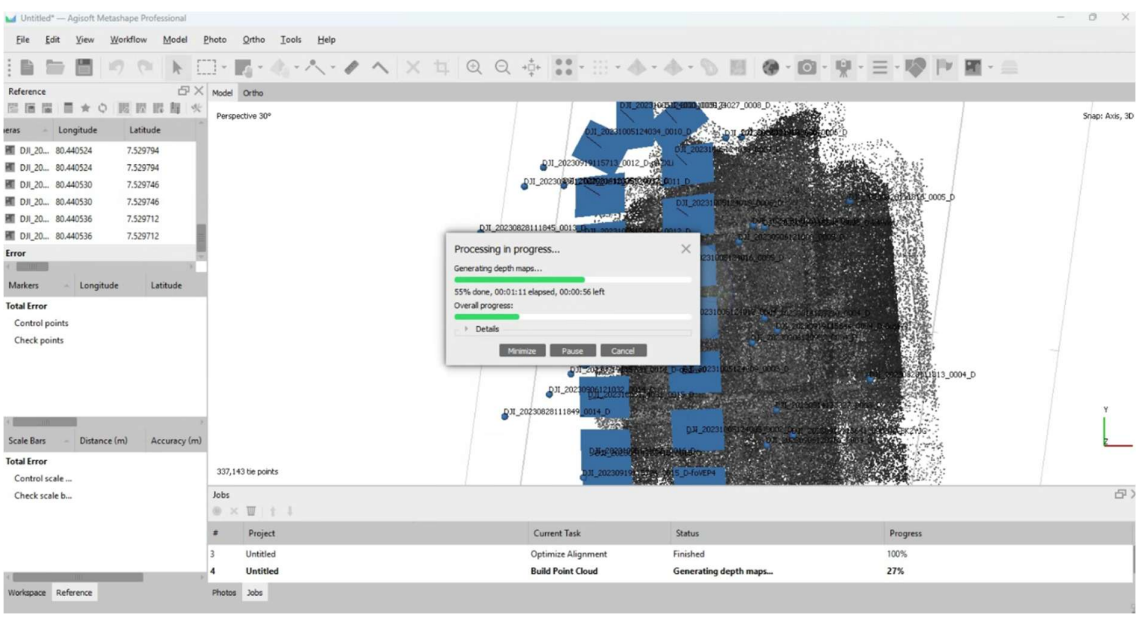


Figure 2: The "Build Point Cloud" task in progress, specifically generating depth maps to create a detailed 3D model.

The screenshot in Figure 2 shows that the "Optimize Alignment" task has finished and the "Build Point Cloud" task is active. The point cloud visible in the main model view is significantly more detailed than the sparse cloud, providing the raw unstructured 3D data necessary for creating a high-resolution surface model. This dense cloud is the direct input for the Digital Elevation Model generation phase.

3.3 Step 3: Digital Elevation Model (DEM) Generation

The generation of a Digital Elevation Model (DEM) is a critical step that transforms the unstructured 3D point cloud data into a structured raster grid. A DEM represents the ground surface elevation, providing an essential output for topographical analysis, hydrological modeling, and, crucially for this workflow, serving as the geometric foundation for orthorectification.

The process is initiated using the "Build DEM" task, which interpolates the elevation data from the dense point cloud onto a regular grid.

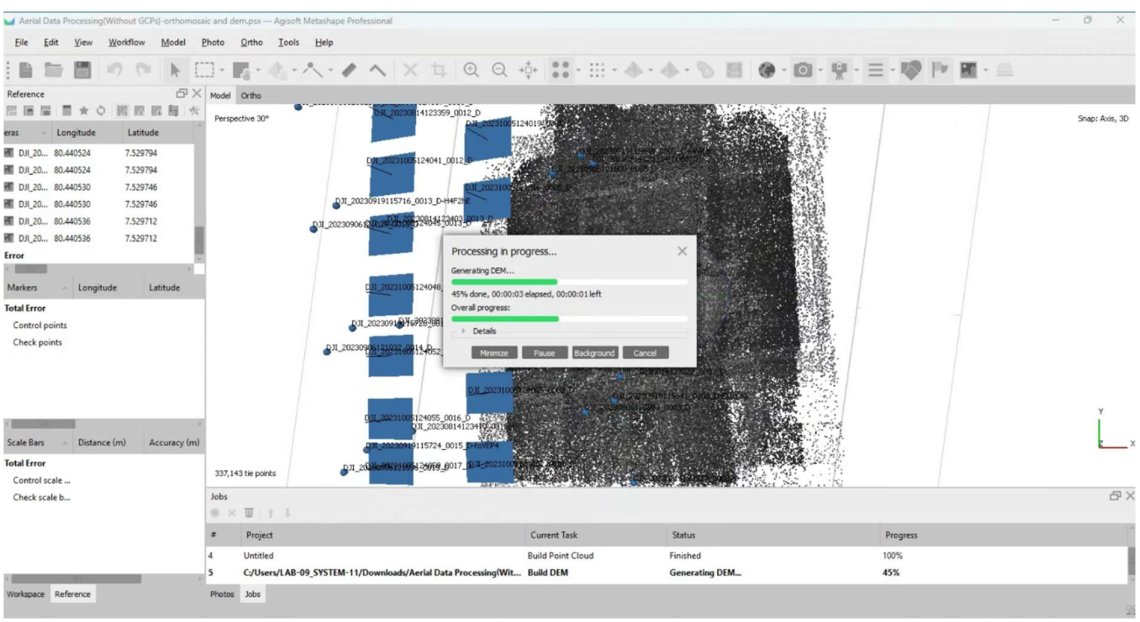


Figure 3: The "Build DEM" task processing the dense point cloud.

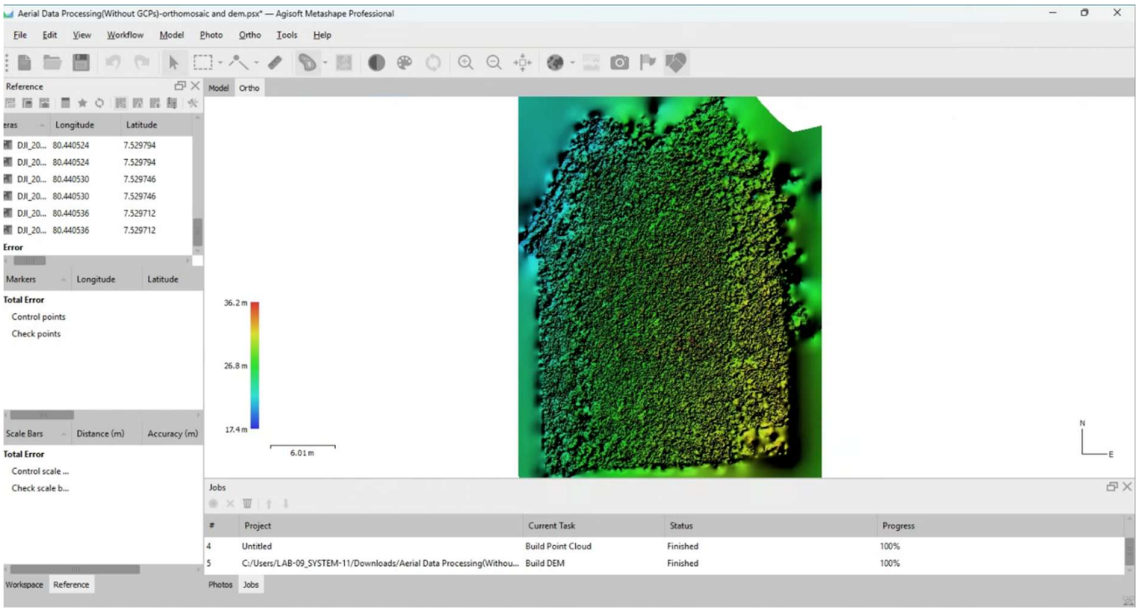


Figure 4: The completed Digital Elevation Model, with elevation values color-coded according to the scale bar.

As shown in Figure 4, the **Jobs** panel confirms that the "Build DEM" task is finished. The resulting DEM is displayed in the main window, with a color ramp visualizing the topographic variation across the project area. According to the scale bar, the elevation ranges from a low of 17.4 meters to a high of 36.2 meters. This completed DEM is now ready to be used as the base layer for generating the final orthomosaic.

3.4 Step 4: Orthomosaic Generation

The final major processing step is the generation of the orthomosaic. An orthomosaic is a seamless, georeferenced aerial image with a uniform scale, free from the distortions caused by camera tilt and terrain variation. The DEM created in the previous step is essential for this process, as it provides the elevation data needed to correct for topographic displacement—a geometric distortion where taller objects appear to lean away from the center of each photo. This operation is known as orthorectification.

The task is initiated with the "Build Orthomosaic" command. The initial status, "Analyzing DEM...", indicates that the software is referencing the elevation model to calculate these corrections for each source image before stitching them together.

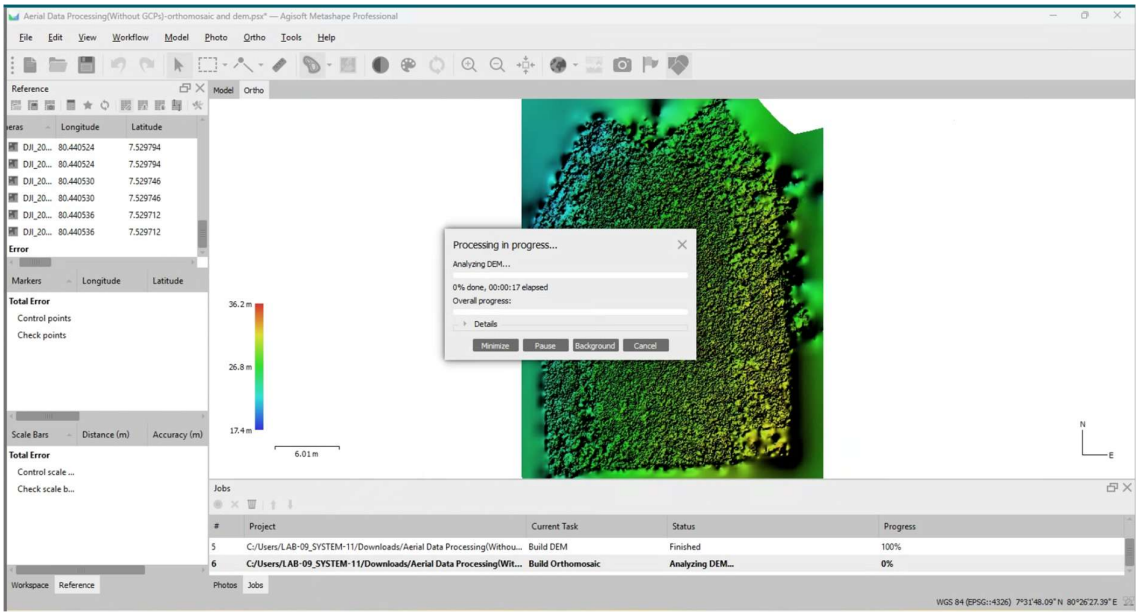


Figure 5: The "Build Orthomosaic" process begins by analyzing the DEM to prepare for image draping and correction.

The interface in Figure 5 confirms that the "Build DEM" task is complete and "Build Orthomosaic" has commenced. During this stage, the software effectively "drapes" and blends the original aerial photographs over the 3D surface defined by the DEM, creating a single, map-accurate image. The final output will be georeferenced to the project's coordinate system, in this case **WGS 84**. The successful completion of this step marks the end of the primary processing workflow.

4.0 Conclusion and Next Steps

This standard operating procedure outlines the four fundamental steps for processing aerial imagery in Agisoft Metashape when GCPs are not utilized: **Photo Alignment**, **Dense Point Cloud Generation**, **DEM Generation**, and **Orthomosaic Generation**. Completing this workflow yields two primary data products—a Digital Elevation Model and a high-resolution orthomosaic—that are valuable for visualization and preliminary analysis.

With these outputs generated, several subsequent analyses and processing tasks can be performed to extract further value from the data. Potential next steps include:

- **Performing DEM-based measurements**, such as calculating stockpile volumes or generating terrain profiles and cross-sections.

- **Classifying the dense point cloud** to algorithmically separate ground points from non-ground objects like vegetation and buildings, which is a prerequisite for creating a Digital Terrain Model (DTM) that represents the bare earth.
- **Generating a textured 3D model** for advanced visualization, stakeholder communication, and detailed structural analysis.